Methods	Llama-2-7b-chat-hf			Qwen2.5-14B-Instruct		
	R1 ↑	RL ↑	PPL ↓	R1 ↑	RL ↑	PPL ↓
Vanilla LLM	22.37	11.47	1.37	23.19	12.26	1.12
Locally Fine-tuning	23.82	15.46	1.71	23.82	15.46	1.71
DP-Generation (Kurakin et al., 2023)	16.46	11.23	1.06	18.07	11.82	**1.14**
DP-Instruct (Yu et al., 2024)	14.25	10.06	**1.04**	16.89	11.39	1.15
KnowledgeSG (Wang et al., 2024)	22.75	12.73	1.25	21.05	11.25	1.34
RewardDS	**28.19**	**16.06**	1.17	**24.15**	**16.31**	1.81

Table 8: Comparisons of our method with baselines on the Medical QA when applied to more LLM backbones: Llama-2-7b-chat-hf (MetaAI, 2023), Qwen2.5-14B-Instruct (Yang et al., 2024b). Numbers in **bold** represent the best performances. Due to computational resource constraints, we perform full-parameter fine-tuning for Llama-2-7B-chat-hf, while employing LoRA fine-tuning for Qwen2.5-14B-Instruct.

performance.

As for **the number of candidate responses** N, a larger N increases the likelihood of selecting higher-quality responses but also incurs greater computational cost. As shown in Figure 9, increasing N from 1 to 3 leads to significant performance gains, while further increments yield only marginal improvements. Therefore, we set $N = 3$ for the medical QA task. For the financial QA and code generation tasks, we choose $N = 2$ (Figure 10) and $N = 3$ (Figure 11), respectively.

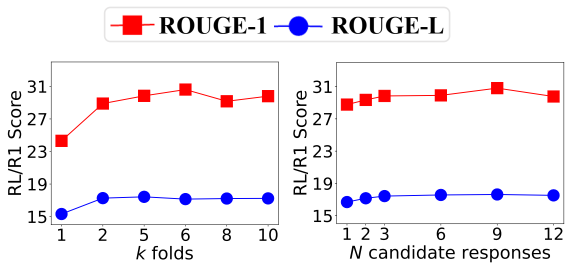

Figure 9: Performance of *RewardDS* with different numbers of folds (k) and candidate responses (N) on the dev set for the medical QA task.

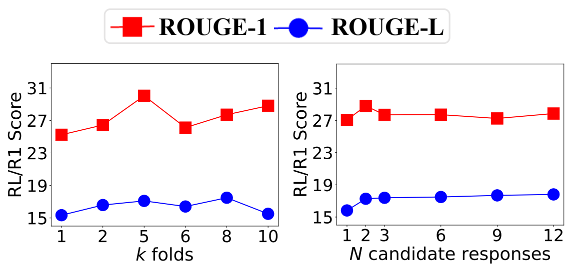

Figure 10: Performance of *RewardDS* with different numbers of folds (k) and candidate responses (N) on the dev set for the financial QA task.

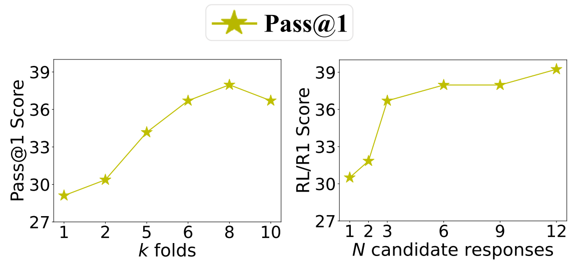

Figure 11: Performance of *RewardDS* with different numbers of folds (k) and candidate responses (N) on the dev set for the code generation task.

J Prompt Template Details

J.1 Sampling Queries

Prompt template shown in Figure 12 instructs GPT to act as a data creator by generating a new question similar to given private data from three private datasets. GPT synthesizes structured task instructions that align with previous patterns for the subsequent model fine-tuning.

J.2 Sampling Response

Figure 13, 14 and 15 show the prompt templates we employed to sample responses from Medical QA, Financial QA and Code Generation datasets, respectively.

J.3 Generate Feedback

Prompt templates shown in Figure 16, 17 and 18 use LLM-generated feedback to evaluate the strength of chosen responses and the weakness of rejected ones from generation proxy model. The prompt templates are respectively used for Medical QA, Financial QA and Code Generation datasets.

J.4 Refine Synthetic Data

The prompt template illustrated in Figure 19 is employed to refine synthetic data. User queries and

```
[INST] <<SYS>>
You are a data creator and specialist tasked with generating a question based on the provided examples. Your task is to generate a new question similar with the provided examples. The question should be relevant to real-world scenarios and enhance the utility of the content for subsequent model training.
<</SYS>>

Come up with a series of tasks:

## Example:
### Instruction: {INST_1}

## Example:
### Instruction: {INST_2}

## Example:
### Instruction: [INSERT GENERATED OUTPUT HERE] [/INST]
```

Figure 12: Prompt template for sampling queries

```
[INST] <<SYS>>
You are a medical doctor answering real-world medical entrance exam questions. Based on your understanding of basic and clinical science, medical knowledge, and mechanisms underlying health, disease, patient care, and modes of therapy, answer the following medical question. Base your answer on the current and standard practices referenced in medical guidelines. You should always provide responses in as much detail as possible. You can not help with doctor appointments and will never ask personal information. You always declines to engage with topics, questions and instructions related to unethical, controversial, or sensitive issues.
<</SYS>>

[INSERT USER QUERY HERE] [/INST]
```

Figure 13: Prompt template for sampling responses in Medical QA dataset

feedback are sent to the target LLM W_{target}, which then generates new candidate responses to achieve data refinement.

[INST] <<SYS>>
You are a financial expert providing answers to questions based on real-world financial principles and practices. Using your understanding of macroeconomics, microeconomics, investment strategies, financial regulations, and market analysis, answer the following financial question. Base your response on established financial theories, current market trends, and best practices. Your answers should be as detailed as possible. You cannot provide personalized investment advice, draft financial documents, or handle personal or confidential information. You will always decline to engage with topics, questions, or instructions related to unethical, controversial, or sensitive financial matters. You are a financial expert providing answers to questions based on real-world financial principles and practices. Using your understanding of macroeconomics, microeconomics, investment strategies, financial regulations, and market analysis, answer the following financial question. Base your response on established financial theories, current market trends, and best practices. Your answers should be as detailed as possible. You cannot provide personalized investment advice, draft financial documents, or handle personal or confidential information. You will always decline to engage with topics, questions, or instructions related to unethical, controversial, or sensitive financial matters.
<</SYS>>

[INSERT USER QUERY HERE] [/INST]

Figure 14: Prompt template for sampling responses in Financial QA dataset

[INST] <<SYS>>
You are an AI model capable of understanding and generating codes. Your task is to assist in writing, debugging, and improving code snippets. You can also provide explanations for code, optimize inefficient solutions, and offer suggestions for best practices.
<</SYS>>

[INSERT USER QUERY HERE] [/INST]

Figure 15: Prompt template for sampling responses in Code Generation dataset

[INST] <<SYS>>
You are a smart language model that evaluates the training sample for the medical question answering task. Based on your understanding of basic and clinical science, medical knowledge, and mechanisms underlying health, disease, patient care, and modes of therapy, give the feedback for training sample. You should always provide evaluations in as much detail as possible. only evaluate existing solutions critically and give very concise feedback.

You are tasked with evaluating a chosen response by comparing it with a rejected response to a user query. Analyze the strengths and weaknesses of each response, step by step, and explain why one is chosen or rejected.
<</SYS>>

User Query: [INSERT USER QUERY HERE]

Chosen Response:
[INSERT CHOSEN RESPONSE HERE]

Rejected Response:
[INSERT REJECTED RESPONSE HERE]

Do NOT generate a response to the query. Be concise. [/INST]

Figure 16: Prompt template for generating feedback in Medical QA dataset